

## Week 5: More induction

$$\forall n \in \mathbb{Z}_{\geq 0} \quad P(n)$$

Base case:  $P_{\text{base}} \quad P(1)$

Inductive Step:  $P_{\text{base}} \quad P(n) \Rightarrow P(n+1)$

ok  $\forall 0 \leq n \leq \infty \quad P(a) \Rightarrow P(a+1)$

$$P(n+1) \Rightarrow P(n)$$

$$P(n) = "2^0 + 2^1 + 2^2 + \dots + 2^{n-1} = 2^n - 1"$$

Want:  $\forall n \in \mathbb{Z}_{\geq 0}, P(n)$

Proof: Proceed by induction. Tk base case

$P(1)$  is " $2^0 = 2^1 - 1$ ", i.e.,  $1 = 1$  which is true.

Assume  $P(n)$  is true. I.e.,

$$2^0 + 2^1 + \dots + 2^{n-1} = 2^n - 1.$$

(WTS:  $P(n+1)$ , i.e.,  $2^0 + 2^1 + \dots + 2^{(n+1)-1} = 2^{n+1} - 1$ )

Then, adding  $2^n$  to each side gives

$$\begin{aligned} 2^0 + 2^1 + \dots + 2^{n-1} + 2^n &= 2^n - 1 + 2^n \\ &= 2(2^n) - 1 \\ &= 2^{n+1} - 1. \quad \square \end{aligned}$$

Let's try  $P(n-1) \Rightarrow P(n)$ .

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$$\begin{aligned} 2^0 + 2^1 + \dots + 2^{(n-1)-1} + 2^{n-1} &= 2^{(n-1)} - 1 + 2^{n-1} \\ &= 2(2^{n-1}) - 1 \\ &= 2^n - 1 \end{aligned}$$

Define:

$$a_0 = 0$$

$$a_n = \sqrt{3 + 2a_{n-1}}$$

$$a_1 = \sqrt{3 + 2a_0} = \sqrt{3+0} = \sqrt{3}$$

$$a_2 = \sqrt{3 + 2a_1} = \sqrt{3 + 2\sqrt{3}}$$

$$a_3 = \sqrt{3 + 2\sqrt{3 + 2\sqrt{3}}}$$

Claim:  $\forall n \in \mathbb{Z}_{\geq 0},$

$$a_n < 3.$$

$$P(n) = "a_n < 3"$$

Proof: Base case:  $P(0)$  is the statement

" $a_0 < 3$ ", i.e., " $0 < 3$ ", which is true.

Suppose that  $P(n-1)$  is true. I.e.,  $a_{n-1} < 3$ .

Then by definition,  $a_n = \sqrt{3 + 2a_{n-1}}$ .

Combining these gives

$$a_n = \sqrt{3 + 2a_{n-1}} < \sqrt{3 + 2 \cdot 3} = \sqrt{3 + 2 \cdot 3} = \sqrt{9} = 3. \quad \square$$

Simplify

$$a_{n-1} < 3$$

$$a_n = \sqrt{3 + 2a_{n-1}}$$

$$< \sqrt{3 + 2(3)}$$

$$= \sqrt{3 + 6} = \sqrt{9} = 3$$

"Any time something works for 2 things,  
it works for many things."

Example: we know that if  $a, b$  are odd, then  
 $ab$  is odd.

$$(a = 2k+1, b = 2m+1, ab = (2k+1)(2m+1) = 4km + 2(km) + 1) \\ = 2(2km + km) + 1)$$

Lemma: let  $a_1, \dots, a_n$  be odd integers. Then

$a_1 a_2 \dots a_n$  is odd.

Proof:  $P(1)$  is just "if  $a_1$  is odd, then  $a_1$  is odd."

$P \Rightarrow P$

$P(2)$  is true.

Proof: we already know  $P(1)$  and  $P(2)$ .

needed 2 base cases

Assume  $P(n-1)$  is true. Let  $a_1, \dots, a_n$  be odd integers

$$(a_1 a_2 \dots a_n = (a_1 a_2 \dots a_{n-1}) \cdot a_n) \\ \text{by } P(n-1), \text{ this is odd.} \\ \text{by } P(n-1), \text{ this is odd.}$$

By induction (i.e., by  $P(n-1)$ ),  $a_1 \dots a_{n-1}$  is odd.

By  $P(2)$  (applied to  $b_1 = (a_1 \dots a_{n-1})$  and  $b_2 = a_n$ )  
 $(a_1 \dots a_{n-1}) a_n$  is odd.  $\square$

Other examples

$$a+b = b+a \quad \text{can rearrange as I wish}$$

$$a_1 + a_2 + \dots + a_n$$

$$= (a_1 + \dots + a_{n-1}) + a_n$$

$$(ab)(c) = a(bc)$$

$$a(a_2 \dots a_n) = (a, a_2)(a_3 \dots a_n)$$

$$(f \circ g) \circ h = f \circ (g \circ h)$$

$$(f_1, f_2, \dots, f_n)$$

Corollary:  $3^a$  is odd  $\forall a \in \mathbb{Z}_{\geq 0}$

$$3^a = (2k+1)^a = \dots$$

binomial expansion

$$P(n) =$$

Prove:  $\forall n \in \mathbb{Z}_{\geq 0}, 3 \mid 4^n - 1.$

Base Case:  $P(0)$ , i.e.,  $3 \mid 4^0 - 1$ , i.e.,  $3 \mid 0.$

This is true. ( $0 = 3 \times 0$ , i.e.,  $x=0$ )

Assume  $P(n)$  is true, i.e.,  $3 \mid 4^n - 1.$

(WTP:  $3 \mid 4^{n+1} - 1$ ) Then  $\exists m \in \mathbb{Z}$  s.t.

$$4^n - 1 = 3m. \quad (4^{n+1} = 4^n \cdot 4) \quad \text{Then } 4^n = 3m + 1.$$

$$\begin{aligned} \text{Thus, } 4^{n+1} - 1 &= 4^n \cdot 4 - 1 = (3m + 1)4 - 1 \\ &= (3m)4 + 4 - 1 \\ &= 3m4 + 3. \end{aligned}$$

This is divisible by 3 by 2 out of 3 rule.

$$\underbrace{(4^n - 1)}_{3 \mid} \cdot 4 = \underbrace{4^{n+1} - 4}_{3 \mid} = \underbrace{(4^n - 1)}_{3 \mid} \cdot \underbrace{3}_{3 \mid}$$

Apply 2 out of 3 rule.

## Fibonacci #'s

$$F_0 = 0$$

0, 1, 1, 2, 3, 5, 8, 13, 21, ...

$$F_1 = 1$$

$$F_2 = 1$$

$$F_3 = 2$$

$$F_{n+2} = F_{n+1} + F_n$$

the duck

## Lucas #'s

$$L_1 = 2, L_2 = 1$$

2, 1, 3, 4, 7, 11, 18, ...

$$L_{n+2} = L_{n+1} + L_n$$

## Confusing Points:

$$F_{a+b} = F_{a+1} + F_a \quad (\forall a)$$

$$F_{n+1} = F_n + F_{n-1} \quad (a = n-1) \quad (n-1)+2 = n+1$$

$$F_n = F_{n+1} + F_{n-2}$$

$$F_{2n} = F_{2n-1} + F_{2n-2}$$

Prove:  $\forall n \in \mathbb{Z}_{>0}, F_1 + F_3 + \dots + F_{2n-1} = F_{2n}.$

Proof: Base case:  $P(1)$  is " $F_1 = F_2$ ", i.e., " $1=1$ ", which is true.

$$(n=2: F_1 + F_3 = F_4, \text{ i.e., } 1+2=3)$$

Assume  $P(n)$  is true, i.e.,

$$F_1 + F_3 + F_5 + \dots + F_{2n-1} = F_{2n}.$$

$$\left( \begin{array}{l} \text{WTD } P(n+1), \text{ i.e., } F_1 + \dots + F_{2(n+1)-1} = F_{2(n+1)} \\ \quad \quad \quad 2(n+1)-1 = 2n+1 \qquad \qquad \quad 2n+2 \end{array} \right)$$

Then, adding  $F_{2n+1}$  to both sides gives

$$F_1 + F_3 + \dots + F_{2n-1} + F_{2n+1} \stackrel{(*)}{=} F_{2n} + F_{2n+1}.$$

Since, by definition,  $F_{2n+2} = F_{2n+1} + F_{2n}$ ,

the RHS of  $(*)$  is the RHS of  $P(n+1)$ .  $\square$

Prove:  $\forall n \in \mathbb{Z}_{>0}, F_n < 2^n$ .

Base cases:

$$P(1) = "F_1 < 2", \text{ i.e., } 1 < 2, \text{ true}$$

$$P(2) = "F_2 < 2^2", \text{ i.e., } 1 < 4. \text{ True}$$

~~Inductive step: assume~~

$$(F_n = F_{n-1} + F_{n-2} < 2^{n-1} + 2^{n-2} < 2^{n-1} + 2^{n-1})$$

$$P(n-1) \wedge P(n-2) \Rightarrow P(n) \quad = 2(2^{n-1}) = 2^n$$

$$P(1), P(2), P(3), P(4), \dots \quad 1 = \text{end}$$



$$F_{n+1} \cdot F_{n+1} = F_n^2 + (-1)^{n+1} \quad P(0)$$

$$P(0) = "F_0 F_2 = F_1^2 + (-1)^1"$$

$$"0 \cdot 1 = 1^2 + (-1) = 0"$$

$$P(1) = "F_1 F_3 = F_2^2 + (-1)^2"$$

$$1 \cdot 2 = 1^2 + (-1)^2$$

Assume  $P(n)$  is true, i.e.,

$$F_{n+1} F_{n+1} = F_n^2 + (-1)^{n+1}$$

$$(WTP \ P(n+1), \text{ i.e., } F_n F_{n+2} = F_{n+1}^2 + (-1)^{n+2})$$

$$F_{n+1} + F_n = F_{n+2} \Rightarrow F_{n+1} = F_{n+2} - F_n$$

$$(F_{n+2} - F_n) F_{n+1} = F_n^2 + (-1)^{n+1}$$

$$F_{n+1}^2 - F_n F_{n+1} =$$

$$=$$

$$F_{n+1}^2 - (-1)^{n+1} = F_n^2 + F_n F_{n+1}$$

$$F_{n+1}^2 + (-1)^{n+2} = F_n(F_n + F_{n+1}) = F_n F_{n+2} \quad \square$$